

Please check the examination details below before entering your candidate information

Candidate surname						Other names					
Centre Number						Learner Registration Number					
Pearson BTEC Level 3 Nationals Extended Certificate, Foundation Diploma, Diploma, Extended Diploma											
Friday 16 May 2025											
Afternoon (Time: 2 hours)						Paper reference		31760H			
Information Technology UNIT 1: Information Technology Systems											
You do not need any other materials.										Total Marks	

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The total mark for this paper is 90.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

- 1** Nino works from home for an insurance company. Nino uses a laptop computer to store large confidential data files.

- (a) Nino sends electronic files to clients. Before the files are sent, they need to be compressed.

Give **two** types of compression.

(2)

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- (b) Nino connects the laptop to his home network using a wired connection.

- (i) Explain **two** advantages to Nino of using a wired connection rather than a wireless connection.

(4)

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(ii) Explain **two** disadvantages to Nino of using a wired connection rather than a wireless connection.

(4)

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- (c) Nino uses his laptop when travelling to meet clients. He uses the mobile data connection on his smartphone to connect to his company's network.

Explain **two** drawbacks to Nino of using a mobile data connection.

(4)

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- (d) Nino needs to backup his files daily. He is considering either an external hard-disk drive (HDD) or an external solid-state drive (SSD).

Discuss the characteristics of HDD and SSD external storage devices for backing up his files.

(6)

(Total for Question 1 = 20 marks)



2 A college runs courses on how to build desktop computers and networks.

(a) Students are taught about operating systems.

Explain **one** factor that affects the performance of an operating system.

(2)

(b) Desktop computers are connected to a range of peripheral devices.

Describe how the operating system manages peripheral devices.

(4)



(c) Students have installed software to protect data.

(i) Explain **two** types of software that can be installed to safeguard against threats to data.

(4)

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(ii) Explain **two** impacts of threats to data, information and systems on the college.

(4)

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(d) Operating systems manage various aspects of computer systems.

Discuss the role of the operating system in managing:

- networking
- security
- memory management
- multitasking.

(8)



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(Total for Question 2 = 22 marks)



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QUESTION 3 BEGINS ON THE NEXT PAGE.



- 3 A family-run business grows fruit and vegetables to supply local supermarkets.

The business has a main office and many greenhouses.

Each greenhouse has a data logging system that automatically monitors and controls the temperature.

The main office has a database server, three desktop computers and a printer.

All the data collected from the data logging systems are stored automatically on the database server in the main office.

The computers, smartphones and tablets in the main office are used to access data and control the temperatures.

Main office

Servers

Desktop computers

Printer

Greenhouses

Data logging system

Mobile devices

Tablets

Smartphones

- (a) Draw a diagram to show how the main office computers, data logging systems and mobile devices interact.

The diagram should include:

- devices and systems that could be used in the main office
- data logging devices in greenhouses
- mobile devices used by the staff
- connection types to be used
- the flow of data through the system
- annotations indicating the information and data to be passed between systems, devices and users.



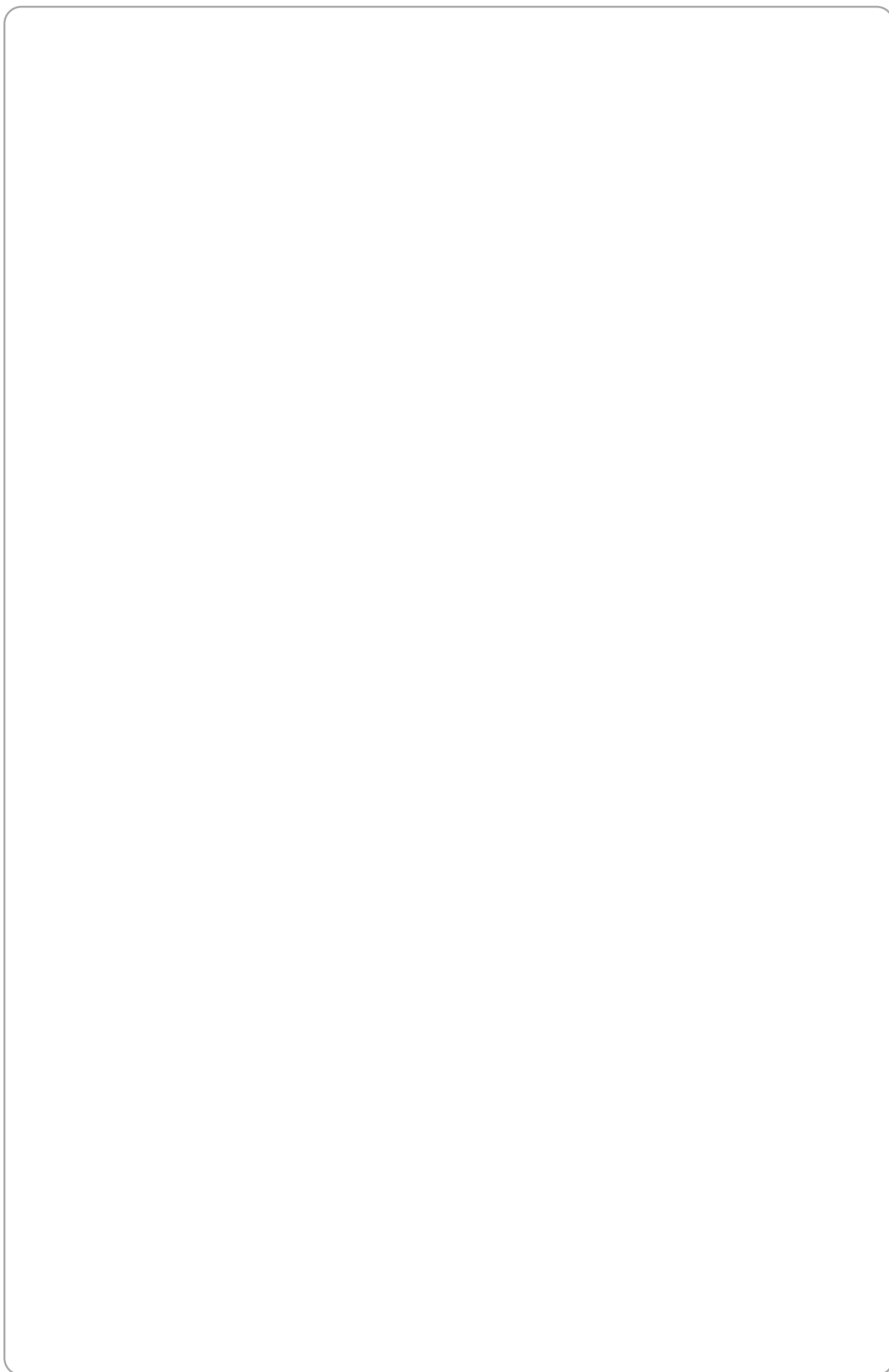
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Use this space to draw your diagram for Q3a.

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(b) Describe how a data logging system works to monitor and control the temperature in a greenhouse.

(4)

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(c) Explain **two** benefits of using a data logging system to monitor and control temperature in the greenhouses.

(4)

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(d) The business analyses data captured from the data logging system.

One reason for the analysis is to help to control the temperature in each greenhouse.

Explain **two other** reasons why the business would analyse the captured data.

(4)

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4 A company sells caravans and motorhomes.

The business creates information packs for the caravans and motorhomes, and makes them available online.

Each sales pack includes the specification, photographs and videos of the caravan or motorhome.

- (a) Discuss the factors that should be considered when making the sales packs available online.

(10)

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- (b) The company wants to set up an online community and forum. Customers will be able to ask questions and share experiences of buying and using a caravan or motorhome.

Discuss how the company can ensure the community members find the user interface easy to use and feel safe while participating.

(12)



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TOTAL FOR PAPER = 90 MARKS

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